

Past Solution #1



This solution is from a science forum called [Science Direct](#), in which they design and build a bear feeder. It looks to work perfectly and is easy to automate. The issue comes when the moat is taken into account, this design would not work to shoot food over a long distance.

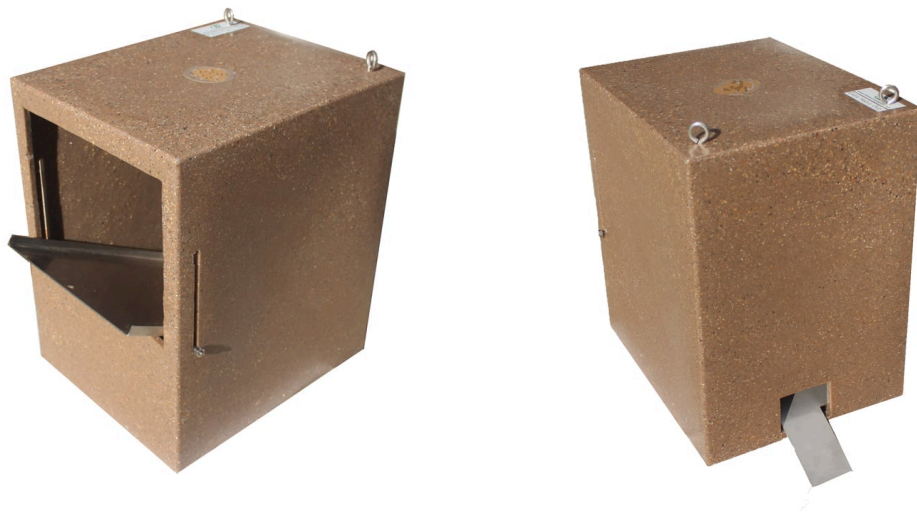
Pros:

1. The automated part of the design is well-documented and seems to work effectively
2. The design is easy to use and doesn't require any extensive technical knowledge
3. The design uses a stepper motor to randomly shoot at different angles; this randomness is exactly what is required in our project.

Cons:

1. The design shoots in a 360 degree angle and needs to be hanging from the middle of the enclosure
2. Can only release food in an area around the feeder; can't shoot far distance

Past Solution #2



A solution given by [Otto Environmental](#), a company specializing in working with animals, is a box that hangs on the edge of a bear enclosure. The box has an adjustable opening from 30 to 60-degree angles. This is what allows them to control where the food goes. Although they did not include an automation function, we can take the idea of a different angle to automate where the feeder will shoot. If we can adjust the angle of the feeder we would accomplish a whole ½ of our criteria.

Pros:

1. Simple to code a sliding door and wouldn't be hard for the build team to make

2. The design itself is simple. The parts are easy to order and put together, it only requires some stainless steel, hooks, and cardboard.

Cons:

1. Doesn't fit our criteria for launching, and doesn't have a good solution for automating the timing of the launch

What we learned

By looking at past solutions for similar situations we were able to see what was applicable to our project and what wasn't. Something that both of these solutions used was the changing of the angle of the launcher. In solution one they do this with a motor and a hanging base, while in solution two they accomplish this with a sliding door. For automating the launching position of the feeder, solution one is more convenient as it has code that we can take and change to be automated. By taking their motor code and changing it to work with our RTC we can time when we want the motor to spin. However, both these designs do not contain any automation of timing for a bear feeder, so we are unable to see if our idea with an RTC will work out well.